

Multiple Choice (10 marks)

1. Complete bilateral cleft palate is due to failure of fusion of the
 - (a) palatine processes.
 - (b) palatine and frontonasal processes.
 - (c) palatine, frontonasal and secondary nasal processes.
 - (d) palatine, frontonasal, secondary nasal processes and mandibular processes.
2. Mapping of human chromosomes:
 - (a) has been restricted to the sex chromosomes because of small family sizes
 - (b) proceeded much more successfully as large numbers of DNA markers became available.
 - (c) has determined that the number of linkage groups is about twice the number of chromosomes
 - (d) has demonstrated that almost all of the DNA is involved in coding for genes
3. The major question concerning aging and personality is:
 - (a) Do older adults still have personalities?
 - (b) Does personality change with age?
 - (c) Does personality decline as we grow older?
 - (d) Does personality increase or decrease with age?
4. Which of the following statements about heritability are true?
 - (a) is a measure of the level of heterozygotes in a population
 - (b) is a measure of the proportion of variation that is due to genetic causes
 - (c) is a measure of proportion of repeated DNA in an organism
 - (d) is a measure of level of gene linkage
5. Fertilization of an ovum by a spermatozoon occurs in which of the following structures?
 - (a) Cervix
 - (b) Fallopian tube
 - (c) Ovary
 - (d) Uterus

True / False (10 marks)

Answer True or False

6. True or False: Androgen insensitivity is a condition that causes female pseudohermaphroditism due to an inability to respond to male hormones.
7. True or False: Older adults may struggle to retrieve memories because they have a larger volume of memories to search through.
8. True or False: Carbon dioxide is the predominant gas found in exhaled air, exceeding the levels of nitrogen and oxygen.
9. True or False: Mutations that cause achondroplasia are classified as gain-of-function mutations, leading to abnormal cartilage development.
10. True or False: Older adults generally perform very well when their working memory is tested.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of togaviruses in infectious diseases and analyze why Yellow Fever is not classified among those caused by this virus family.
12. Discuss the role of various hormones in the aging process, specifically analyzing why vitamin D has not been recognized as a potential agent against senescence.

End of Paper